

SAMEER PANDEY

+91 6204570289 | sameerpandey17nov@gmail.com | Pune, Maharashtra, India | Open to Remote
[LinkedIn](#) | [GitHub](#) | [Twitter / X](#) | [Portfolio](#)

EDUCATION

Dr. D. Y. Patil Institute of Technology — Pune, India
Bachelor of Engineering in Artificial Intelligence & Data Science | Sep 2025 – Present
Second Year | CGPA: 8.71 / 10.0

SKILLS

Languages: Python, TypeScript, SQL
Backend: FastAPI, REST APIs, SQLAlchemy, WebSockets
Frontend: React, Redux Toolkit
Databases: PostgreSQL, SQLite, Redis
Infrastructure: Docker, Docker Compose, Nginx, Redis Pub/Sub
AI / ML: LangGraph, LLM Integration, MediaPipe, OpenAI Gym
Tools: Git

PROJECTS

VisionLink — Real-Time Emotion Detection System

FastAPI, React, PostgreSQL, Redis Pub/Sub, Docker Compose, Nginx, MediaPipe | GitHub

- Architected a real-time full-stack system using FastAPI, Redis Pub/Sub, and WebSockets to process and distribute emotion-detection results across multiple clients and workers.
- Implemented persistent detection history with PostgreSQL, including cursor-based pagination for API responses.
- Added API-key authentication, background database writes, and structured error logging for reliable request processing.
- Containerized the complete application with Docker Compose and configured Nginx as a reverse proxy.
- Built the React frontend with Framer Motion and integrated MediaPipe Face Mesh for six-emotion classification.

AIVOA — AI-Powered Deviation Intake & Copilot

Python, FastAPI, LangGraph, Groq LLMs, React, Redux Toolkit, PostgreSQL/SQLite, Docker Compose | 2026 | GitHub

- Built a full-stack application with FastAPI and React for capturing, editing, and querying pharmaceutical manufacturing deviations.
- Designed a structured processing pipeline using intent routing, LLM extraction, Pydantic validation, and merge/diff logic for controlled field updates.
- Implemented database persistence with SQLAlchemy, PostgreSQL through Docker Compose, and automatic SQLite fallback.
- Added support for PDF, DOCX, XLSX, TXT, and scanned-image uploads with client-side validation and OCR-based processing.
- Validated backend and end-to-end workflows across 10 acceptance scenarios using automated tests and manual browser testing.

CaloRupree — AI Budget Meal Planner

Python, FastAPI, React, TypeScript, PostgreSQL, AI Integration | 2026 | GitHub

- Independently built and deployed a full-stack application using FastAPI, PostgreSQL, React, and TypeScript to generate budget-constrained meal plans.
- Designed API-driven meal planning around user location, ingredients, dietary preferences, calorie targets, protein goals, and daily budget.
- Integrated two AI providers with automatic failover to maintain service continuity during provider outages.
- Built a dashboard for tracking meals, calorie and protein intake, and spending, backed by a 45-meal dataset.

NutriSync — Reinforcement Learning Meal Planning Environment

Python, OpenAI Gym, Groq/LLaMA, Gradio | GitHub

- Designed an OpenEnv-compatible Python environment for sequential meal planning under nutritional and budget constraints.
- Implemented reward calculation with penalties and hard constraints for allergen usage and budget violations.
- Created three difficulty tiers with progressively stricter nutritional and availability constraints.
- Built a 50-ingredient Indian food dataset and Gradio interface for interactive environment exploration.

ACHIEVEMENTS

1st Place — Impromptu Hackathon | 3-Hour Sprint | March 2026

- Won first place by independently building a full-stack college canteen pre-ordering system with real-time menu updates, admin workflows, and payment gateway integration.

RELEVANT COURSEWORK

Data Structures & Algorithms, Database Systems, Computer Networks, Operating Systems, Artificial Intelligence